Ziphozihle Maqpi

88/90

98%

| | |
|---|---|
| **Module #** | 671101000-KM-07: |
| **NQF Level** | level 4, |
| **Notional hours** | 310 |
| **Credit(s)** | Cr 31 |
| **Occupation or Specialisation** | 671101000: Occupational Certificate: Electrician |
| **Associated Occupation** | 671101: Electrician |

# ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

## FORMATIVE ASSESSMENT



Z

| About the Formative Assessment | |
|---|---|
| **Purpose** | The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed. |
| **Context** | This assessment represents the Formative Assessment component and should be completed in the classroom/training room |
| **Resources** | The following are resources needed for this assessment:<br>1. Learner Guide; and<br>2. Assessment Preparation. |
| **Instructions to Facilitators** | Facilitators will be required to:<br>• Explain the completion of the Formative Assessment to each learner; and<br>• Interview the learner on similar questions, should he/she not be able to write. |
| **Instructions to Learners** | Learners will be required to:<br>• Complete the Formative Assessment as per the instructions;<br>• Ensure that all questions are completed;<br>• Ensure that the completion of the workbook is their own work;<br>• Ensure that all annexure are attached to the workbook and clearly referred to; |
| **Assessment Time** | Learners are required to complete this assessment within the allocated time frame/hours. |
| **Total Mark** | This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks. |



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**Special arrangements for assessment**

| |
|---|
| Place |
| Language |
| Resources |
| Barriers |

**People to be involved with assessment**

| | |
|---|---|
| Learner: Ziphozihle Mqopi | Manager: |
| Trainer: | Mentor / Coach: |
| Facilitator | |
| Assessor: | Moderator: |

**Next steps for learning**

**Resources required for this assessment**

**Guidelines to the learner:**

Learners Name: Ziphozihle Mqopi

Learner's signature: _____ Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____



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**Special arrangements for assessment**

| |
|---|
| Place |
| Language |
| Resources |
| Barriers |

**People to be involved with assessment**

| | |
|---|---|
| Learner: | Manager: |
| Trainer: | Mentor / Coach: |
| Facilitator | |
| Assessor: | Moderator: |

**Next steps for learning**

| |
|---|
| |
| |
| |

**Resources required for this assessment**

| |
|---|
| |
| |
| |
| |
| |

**Guidelines to the learner:**

| |
|---|
| |
| |
| |
| |
| |

Learners Name: _____

Learner's signature: _____ Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

**Grounds for Appeal**

| No | Tick the applicable ground(s) for appeal | Tick |
|---|---|---|
| 1 | The assessment did not follow the laid down procedure | |
| 2 | Not all evidence available was taken into account during the assessment | |
| 3 | The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process | |
| 4 | The assessment was not according to the performance criteria and the range statement stipulated in the unit standard | |
| 5 | Not all the range items were available for the assessment | |

## Reasons for Appeal

| No | Please give detailed reasons for the choice(s) above |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____



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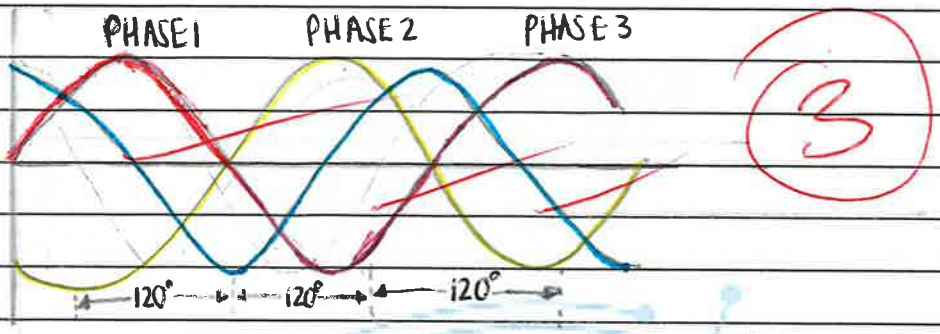
| Learner declaration of acceptance of assessment instruments and relevant documentation: Date: | |
|---|---|
| Learners Name: Ziphozinhle Mqopi | Signature |
| Facilitator Name: | Signature: |

## LEARNING UNIT 1:

## KM-07-KT01: Concepts, theories and principles of supply Systems (20%)

| | |
|---|---|
| **Learning Outcome 1:** | KT0101 Theories and concepts of alternating current |
| **Learning Outcome 2:** | KT0102 Fundamental principles of alternating current |
| **Learning Outcome 3:** | KT0103 Alternating current generation (distribution systems theory) |
| **Learning Outcome 4:** | KT0104 Characteristics and calculations from alternating current waveforms |
| **Learning Outcome 5:** | KT0105 Theories, concepts and principles of direct current sources |
| **Learning Outcome 6:** | KT0106 Sources of direct current |
| **Learning Outcome 7:** | KT0107 Calculations of direct and alternating current circuits |
| **Learning Outcome 8:** | KT0108 Construction and operating principles of direct current generators |
| **Learning Outcome 9:** | KT0109 Characteristics, sources and generation of renewable energy |
| **Learning Outcome 10:** | KT0110 Relevant legislative requirements |

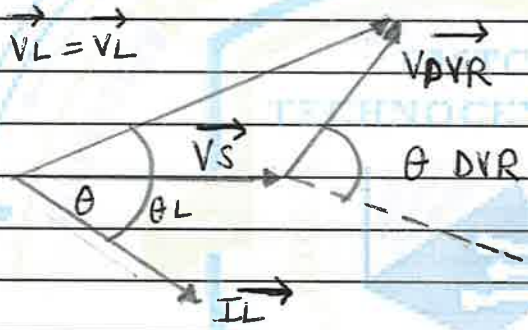
| Task | Questions Description | Marks |
|---|---|---|
| 1. | a) Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams. | 4 |
| | b) Explain by drawing wave form diagrams, the differences between line and phase values. | 4 |
| | c) Explain the generation principles of direct current. | 4 |
| | d) Calculate power and energy in direct and alternating current circuits | 4 |
| | e) Explain with the aid of drawings the operating principles of direct current generators. | 4 |



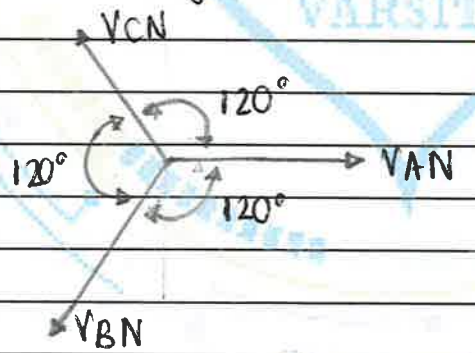

3

3 sinusoidal waveforms of equal amplitude and frequency. Each wave is displaced by 120° electrical from the others

Single phase vector diagram




Three phase vector diagram




$V_{RN} = V_{AN}$

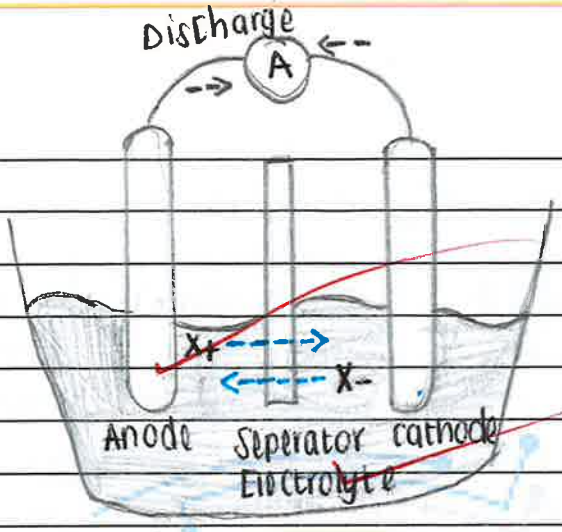
$V_{YN} = V_{BN}$

$V_{BN} = V_{CN}$


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2a)




- Anode (Negative electrode): The negative side of the battery where oxidation occurs. It releases electrons into the external circuit during discharge.
- Cathode (Positive electrode): The positive side of the battery where reduction occurs and electrons are received from the external circuit
- Electrolyte: A chemical medium often liquid/gel that allows the flow of electrical charge between anode and cathode.
- Separator: A porous non-conducting barrier placed between the anode and cathode that physically seperates them to prevent a short circuit, while still allowing ions to pass through.

2b)

| Cell Type | Advantages | Disadvantages | Applications |
|---|---|---|---|
| Primary cell (Non-rechargeable) | • Cheaper to purchase initially<br>• Long shelf life with low self-discharge<br>• Provides instant reliable power without waiting to charge | • Cannot be recharged once depleted it must be discarded<br>• Greater enviromental waste and disposal concern | Used in emergency devices like smoke detectors, TV remotes flashlights and wall clocks |


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| LEARNING UNIT 3:<br>KM-07-KT03: Transformers (20%) | |
|---|---|
| **Learning Outcome 1:** | KT0301 Theories of single phase wound transformers |
| **Learning Outcome 2:** | KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications. |
| **Learning Outcome 3:** | KT0303 Fundamentals of transformer construction |
| **Learning Outcome 4:** | KT0704 Transformer cooling systems |
| **Learning Outcome 5:** | KT0305 Principles of single phase transformer operation |
| **Learning Outcome 6:** | KT0306 Principles of single phase auto-transformer operation |
| **Learning Outcome 7:** | KT0307 Transformer losses |
| **Learning Outcome 8:** | KT0308 Formulas and calculations on input and output of transformers |
| **Learning Outcome 9:** | KT0709 Maintenance requirements of transformers |
| **Learning Outcome 10:** | KT0310 Types of three phase transformers and their applications |
| **Learning Outcome 11:** | KT0311 Construction of three phase transformers including open core, closed core and shell or divided core |
| **Learning Outcome 12:** | KT0312 Principles of 3 phase transformer operation and configuration |
| **Learning Outcome 13:** | KT0313 Principles of 3 phase auto transformer operation and configuration |
| **Learning Outcome 14:** | KT0314 Transformer protection and phasing |

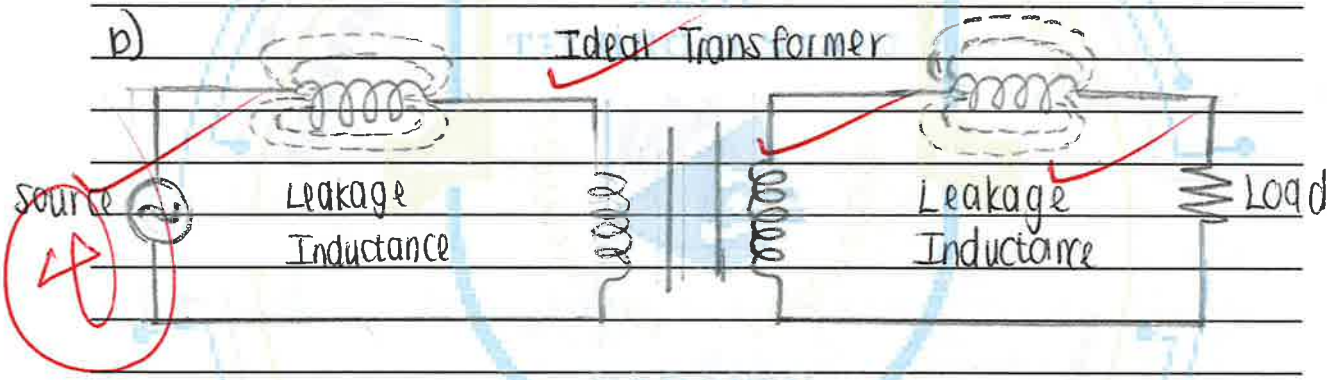

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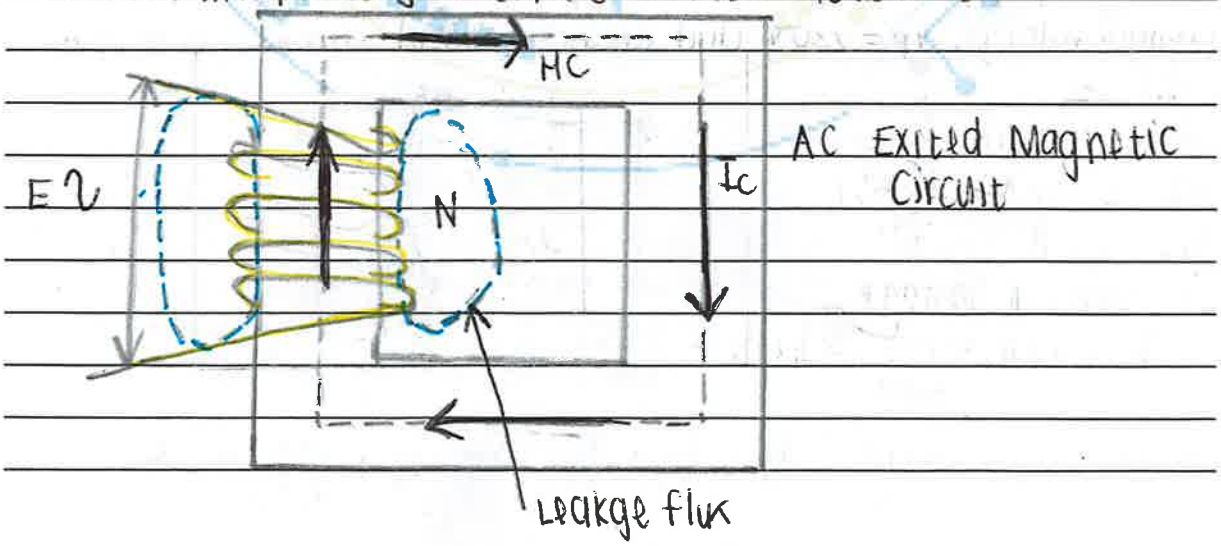
The construction of a single-phase transformer involves:

- The magnetic Core: Built with thin silicon-steel sheets (laminations) coated with varnish to minimize core losses. It consists of vertical limbs where windings are placed and horizontal sections that complete the magnetic circuit
- The Windings: Made of high-grade, enamel-coated copper wire to conduct current efficiently. There are two types: the Primary Winding and Secondary
- Insulation: Thin layers of insulating paper or enamel are placed between the core and the windings and between the coil layers themselves, to prevent electrical short circuits

b)




Mutal Inductance: The time-varying flux links both the primary and secondary windings. The generation of an electromotive force (EMF) in the secondary coil due to changing current in the primary is called mutal inductance.





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## LEARNING UNIT 4:

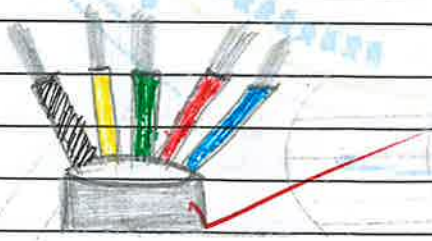
## KM-07-KT04: Types of cables and applications (20%)

| | |
|---|---|
| **Learning Outcome 1:** | KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables. |
| **Learning Outcome 2:** | KT0402 Cable materials and their functions and characteristics |
| **Learning Outcome 3:** | KT0403 Identification of cable characteristics and properties |
| **Learning Outcome 4:** | KT0404 Applications of various types of cable |
| **Learning Outcome 5:** | KT0405 Installation methods and safe use of cables |
| **Learning Outcome 6:** | KT0406 Safe transport and storage of cables |

| Task | Questions Description | Marks |
|---|---|---|
| 4 | a) Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used | 4 |
| | b) Describe the factors effecting the effeciency of cables | 3 |
| | c) Describe the precautions required when transporting and storing cables | 3 |

4a)

1.



PVC

Wire

Construction: Copper and pvc coating

Function and Material: Copper is the main current carrier

PVC insulation: The coloured plastic coating. PVC is cheap, fire-resistant and does not conduct electricity. Each colour is a different wire which are live, neutral and earth.


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## LEARNING UNIT 5:

## KM-07-KT05: Switchgear and control gear (20%)

| | |
|---|---|
| **Learning Outcome 1:** | KT0501 Principles of operation of switchgear and control gear |
| **Learning Outcome 2:** | KT0502 Components of switchgear and control gear systems and the application thereof. |
| **Learning Outcome 3:** | KT0503 Electrical drawings |
| **Learning Outcome 4:** | KT0504 Introduction to Programmable Logic Controllers (PLC's) |
| **Learning Outcome 5:** | KT0505 Introduction to Soft starters |
| **Learning Outcome 6:** | KT0506 Introduction to Variable speed drives |

| Task | Questions Description | Marks |
|---|---|---|
| 5 | a) Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms | 4 |
| | b) Describe the basic principles of operation for programmable logic controllers | 3 |
| | c) Describe the basic principles of operation for variable speed drives | 3 |

5a) Disconnectors

construction: Mechanical switches with visible open contacts, no arc quenching chamber

Operation: operated manually or motor-driven. Used for visible isolation only, must be operated off-load. ~~No~~ Operating coil contacts seperate to provide a ~~safe~~ air gap

4


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2

c) Rectification: The incoming AC supply is converted to DC using a rectifier. This creates a constant DC voltage in the DC bus

Inversion: The DC voltage is switched on and off by an inverter using power electronics like IGBTs (Insulated Gate Bipolar Transistors) to reconstruct an AC output with variable frequency and voltage

Speed control: By varying the frequency and voltage supplied to the motor, the VSD controls the motor's speed and torque according to the load requirements

3


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4

| | |
|---|---|
| Torque: Lower starting torque, pulsating torque to more vibration and noise | Torque: High starting torque uniform or constant torque to smooth, less vibration |

c) Static (Offline) Tests: Using a multimeter or megahmmeter conducted ~~with the power~~ when the motor is unpowered. A multimeter or megger is used for a insulation test to check for winding continuity, correct phase resistance and to verify that the winding insulation has not degraded

Dynamic Tests: Performed while the motor is running under various loads. These measure operating parameters like voltage, running current, temperature rise and vibration to assess overall motor efficiency and indentify mechanical or electrical faults.

2

d) Given: A 3-phase, 415 V, 50 Hz induction motor draws 10 A line current at 0,8 lagging power factor Efficiency = 90%.

Calculate:

1) Input power

$P_{in} = \sqrt{3} \times V_L \times I_L \times \cos\theta$

$P_{in} = 1,732 \times 415 \times 10 \times 0,8 = 5750,24\ W$

2

2) Output power

$P_{out} = P_{in} \times \eta$

$P_{out} = 5,75 \times 0,9 = 5,18\ KW$


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## ASSESSMENT FEEDBACK REPORT

| FACILITATOR FEEDBACK & REMARKS |
| --- |
| |

| ASSESSMENT JUDGEMENT | | |
| --- | --- | --- |
| **Learner's Total Mark:** | **Requirements met** ☐ | **Requirements not met** ☐ |
| **Action/s required:**<br><br>**By when:** | | |

| LEARNER FEEDBACK & COMMENTS |
| --- |
| Good work |

| DECLARATION BY THE FACILITATOR |
| --- |
| I ..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner. |



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## Application for Assessment

First Attempt □ Second Attempt □ Third Attempt□ RPL □

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

## Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

| 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation |
| --- |
| What resources would you require to conduct the assessment?<br>1. Assessor toolkit<br>2. Theoretical assignment<br>3. Practical assessment<br>4. Learner toolkit |
| How would you sequence your activities? Explain<br>1. Record preparation process<br>2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment<br>3. Learner will write summative assessments<br>4. Conduct observation in workplace where applicable<br>5. Complete learner toolkit<br>6. Complete assessor toolkit |



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